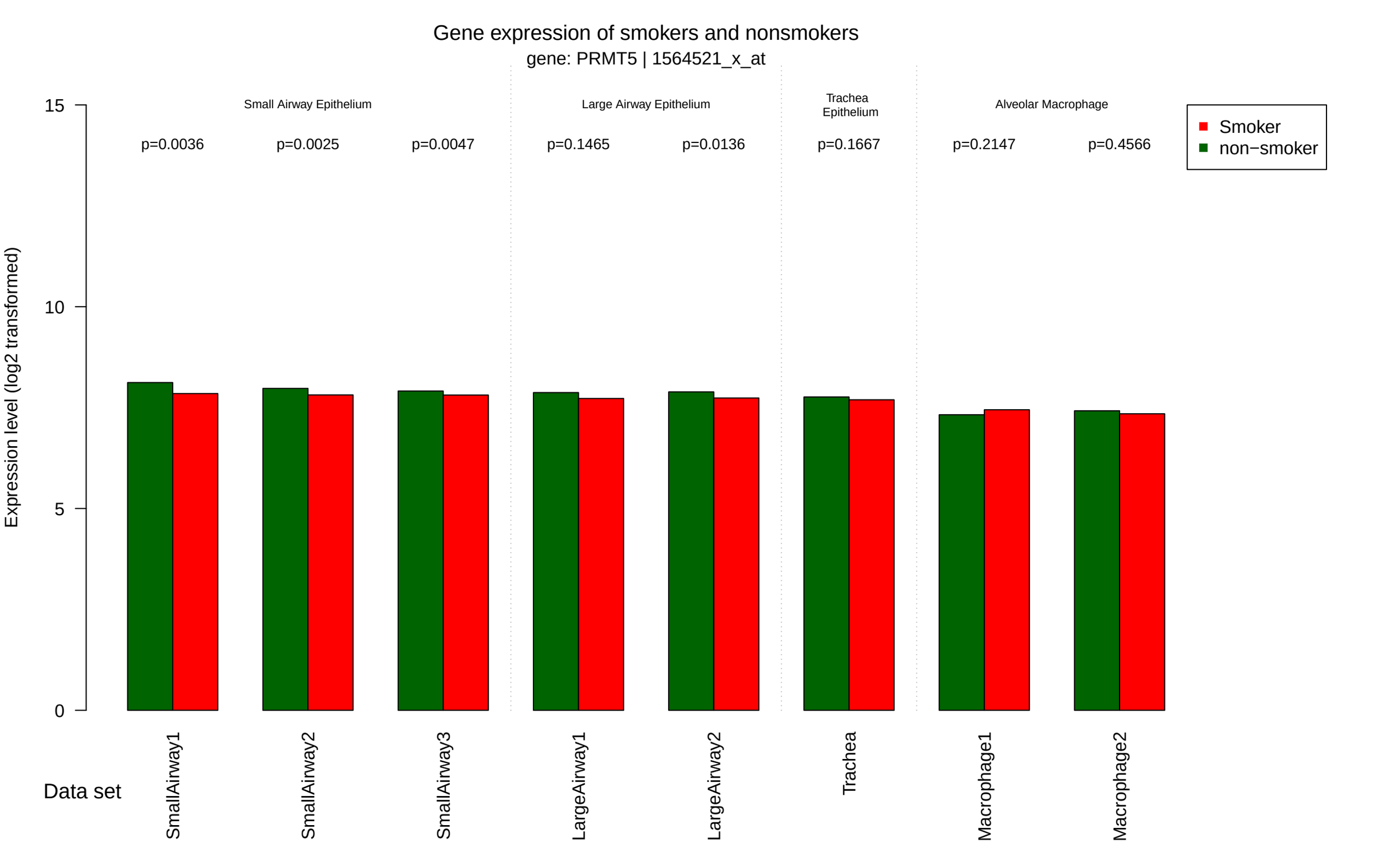
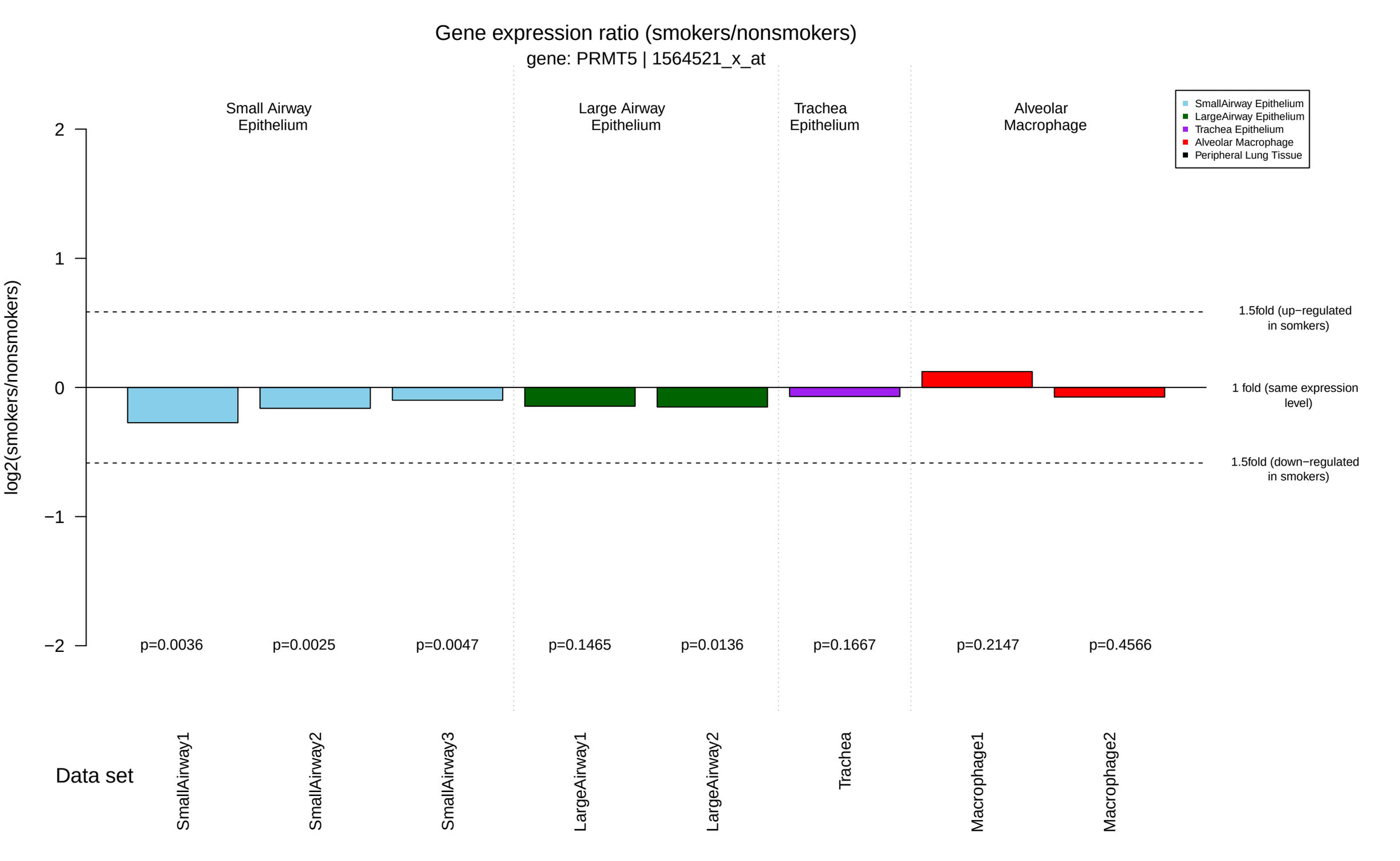
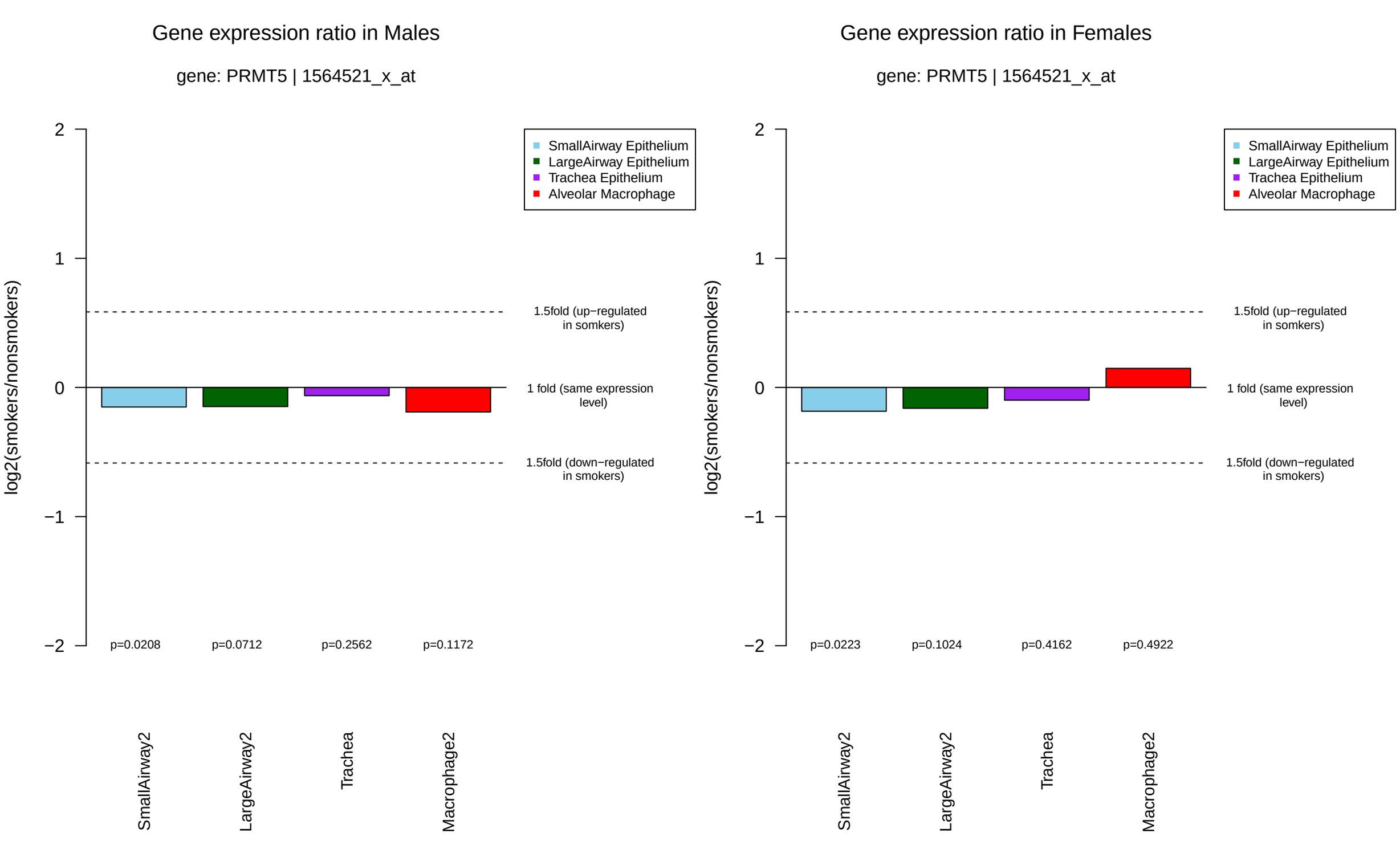
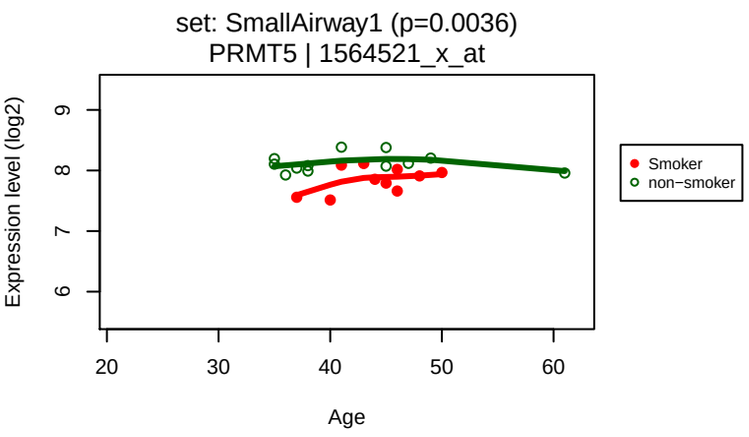
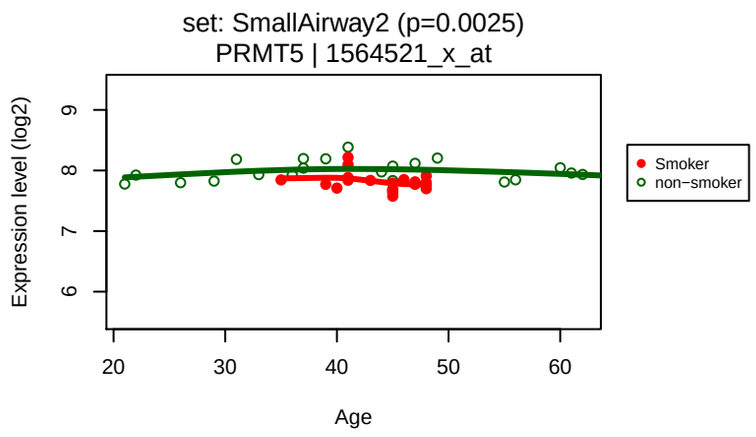
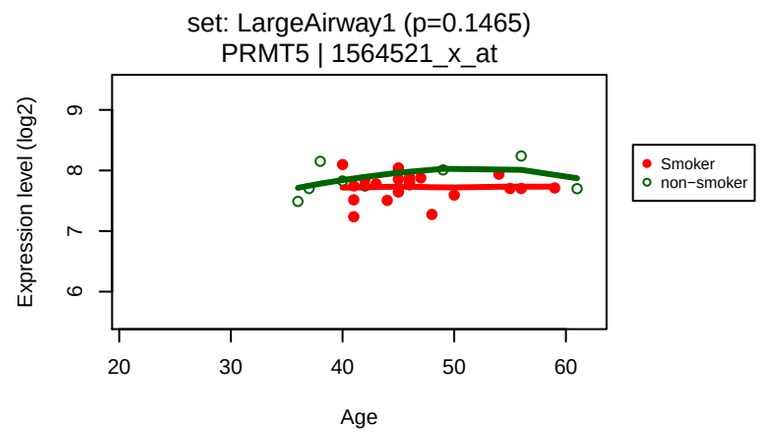

Gene expression of smokers and nonsmokers
gene: PRMT5 | 1564521_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0036
p=0.0025
p=0.0047
p=0.1465
p=0.0136
p=0.1667
p=0.2147
p=0.4566
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

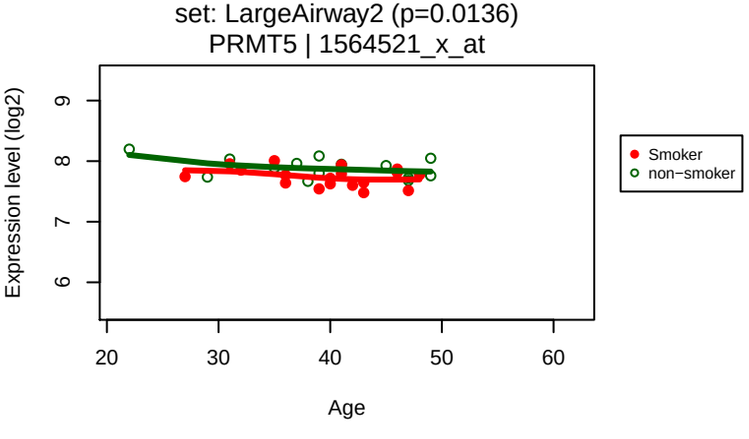

Gene expression ratio (smokers/nonsmokers)
gene: PRMT5 | 1564521_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0036
p=0.0025
p=0.0047
p=0.1465
p=0.0136
p=0.1667
p=0.2147
p=0.4566
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

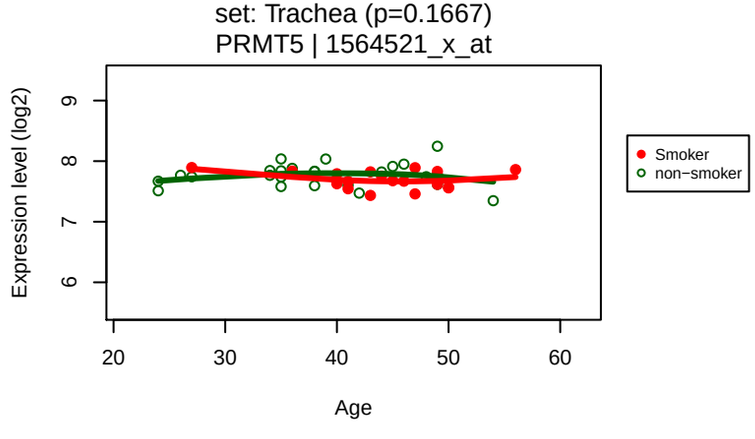

Gene expression ratio in Males
gene: PRMT5 | 1564521_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0208
p=0.0712
p=0.2562
p=0.1172
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRMT5 | 1564521_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0223
p=0.1024
p=0.4162
p=0.4922
SmallAirway2
LargeAirway2
Trachea
Macrophage2

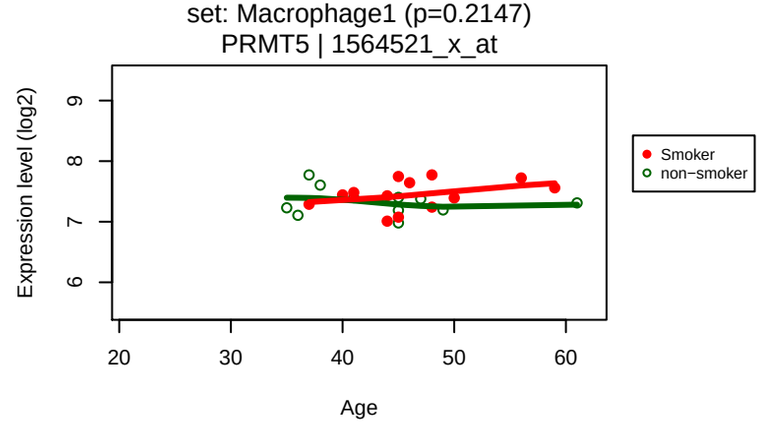

set: SmallAirway1 (p=0.0036)
PRMT5 | 1564521_x_at
Expression level (log2)
Age
Smoker
non−smoker

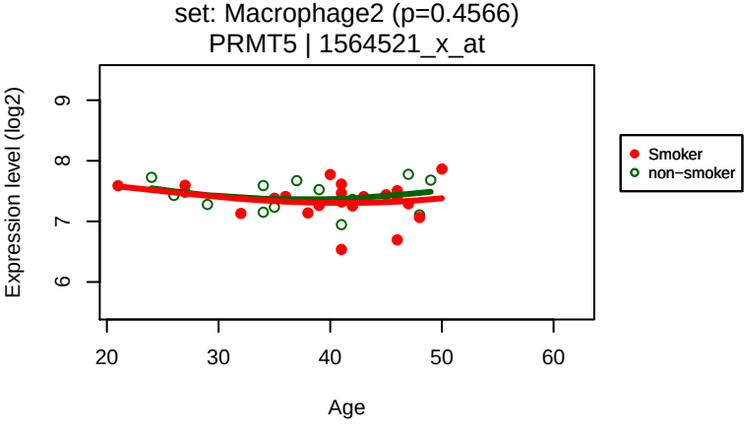

set: SmallAirway2 (p=0.0025)
PRMT5 | 1564521_x_at
Expression level (log2)
Age
Smoker
non−smoker

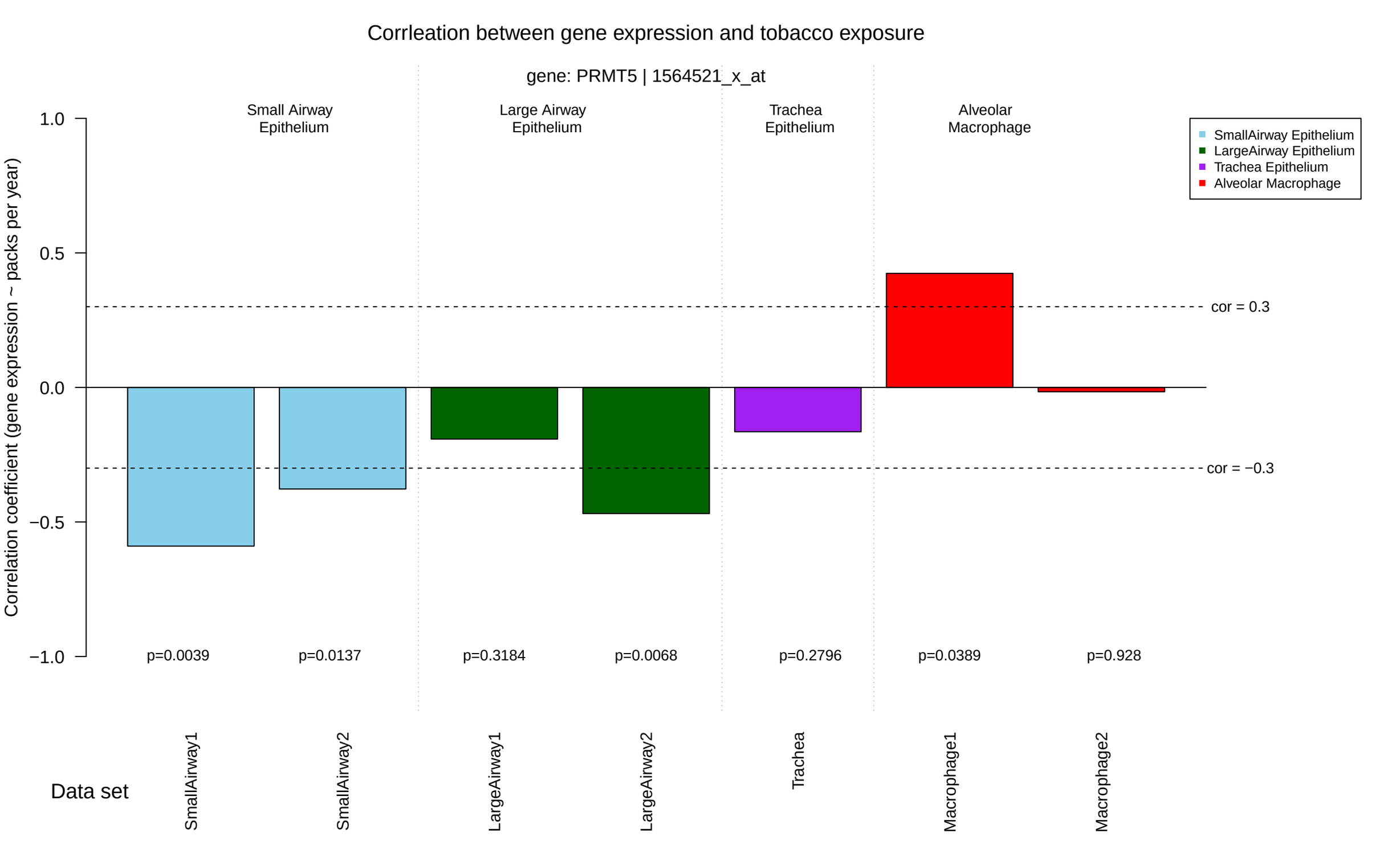

set: LargeAirway1 (p=0.1465)
PRMT5 | 1564521_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0136)
PRMT5 | 1564521_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1667)
PRMT5 | 1564521_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2147)
PRMT5 | 1564521_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4566)
PRMT5 | 1564521_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRMT5 | 1564521_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0039
p=0.0137
p=0.3184
p=0.0068
p=0.2796
p=0.0389
p=0.928
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2